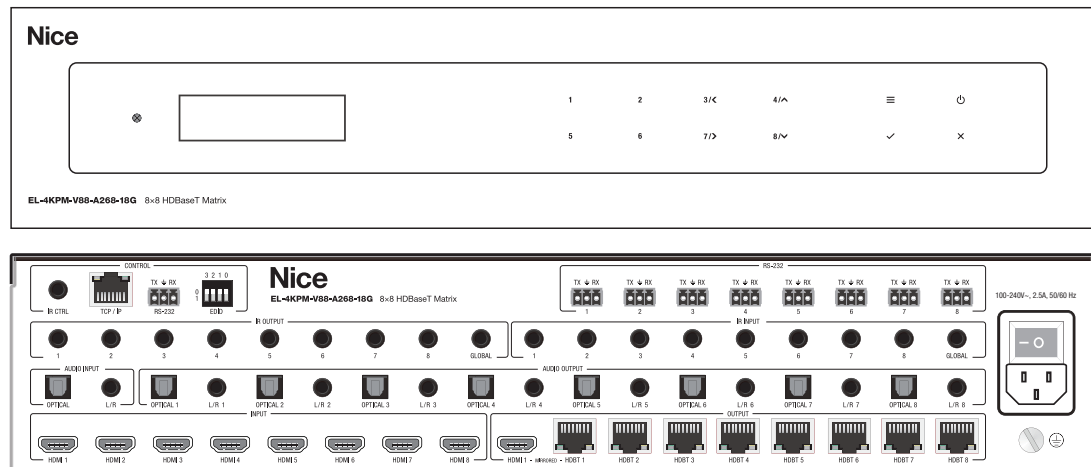


Nice

EL-4KPM-V88-A268-18G

Quick Reference Guide

WORKS WITH ELAN
CONTROL SYSTEMS
ELAN
a Nice brand



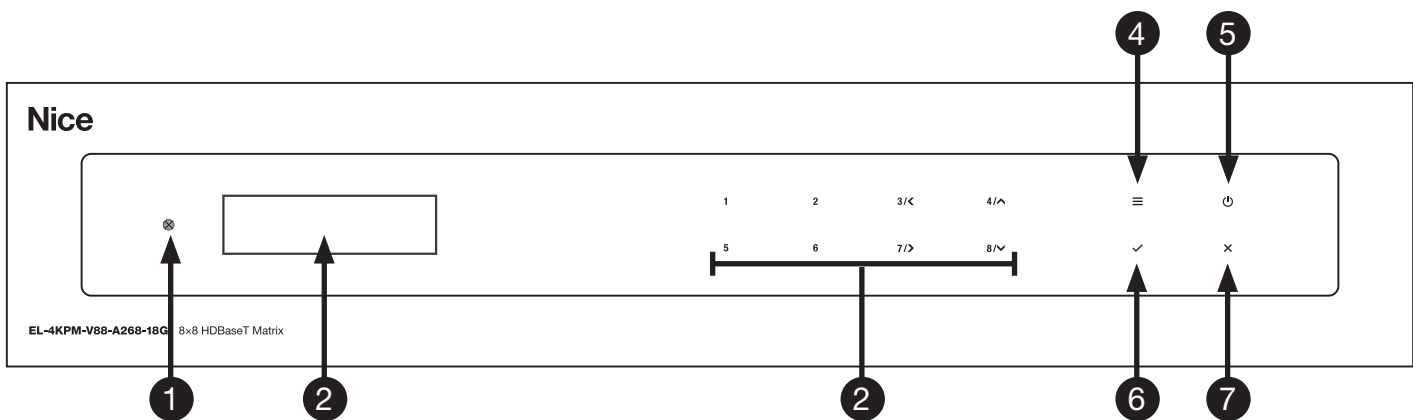
Introduction

The Nice 8x8 HDBaseT Matrix offers unprecedented performance and value for the custom installation market. The EL-4KPM-V88-A268-18G is a HDMI 2.0 4K 60Hz 4:4:4 HDCP 2.2 Matrix package utilizing CSC technology to deliver HDMI, Bi-directional IR and PoC up to lengths of 70m over a single CAT cable. The Matrix also provides advanced features including simultaneous HDBaseT / HDMI on output 1, video down conversion on HDBaseT outputs and independent 26x8 audio Matrix including Audio Return Channel (ARC). The EL-4KPM-V88-A268-18G includes in-built web browser interface module for control and configuration of the Matrix, along with RS-232 and IR pass-through for seamless 3rd party control integration.

Features

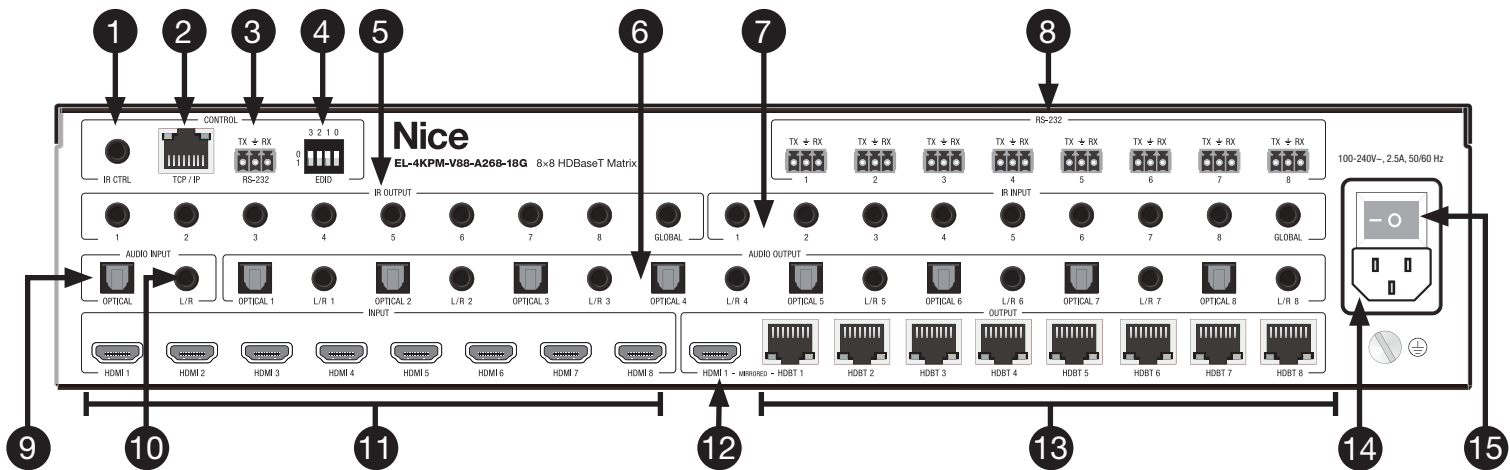
- Advanced HDBaseT technology offering distribution of video and audio over a single CAT cable
- Advanced Color Space Conversion (CSC) supports HDMI 2.0 18Gbps specification including HDR
- Features 8 x HDMI inputs which can be independently routed to 8 x HDBaseT outputs
- Output 1 features simultaneous HDMI and HDBaseT output
- Video down-conversion on HDBaseT outputs allowing a display only capable of supporting lower video resolutions (4K 60Hz 4:2:0, or 1080p) to receive 4K 60Hz 4:4:4 video content, while still showing maximum original 4K UHD resolution on remaining video outputs
- Supports 4K 60Hz 4:4:4 UHD video up to 70m, and 1080p video up to 100m
- 26x8 Audio Matrix independently controllable from video. Audio source inputs include:
 - » 8 x audio breakout from HDMI source inputs
 - » 8 x audio breakout from zone outputs
 - » 8 x ARC from zone outputs*
 - » 1 x Optical and 1 x analogue audio input
- Supports all known HDMI audio formats including Dolby TrueHD, Dolby Atmos, Dolby Digital Plus and DTS-HD Master Audio transmission
- Web interface module for control and configuration of Matrix
- Supports bi-directional IR and RS-232 on all HDBaseT outputs
- Control via front panel, IR, RS-232 and TCP/IP
- Supports PoC (Power over Cable) to power compatible HDBaseT receivers
- Supplied with Nice 5V 3.5mm IR receivers and emitters
- Advanced EDID management and HDCP 2.2 compliant (*ARC feature compatible with EL-4KHDBT-RX-70-ARC-18G only)

Front Panel Description



1. IR Receiver Window	<u>RS-232 control port</u> : Connect to a third party control device for control of the VIP system using RS-232.
2. LCD Display	Shows the status of input / output selection, EDID etc.
3. 'SELECT' Buttons	Multi-use buttons. Primary Use: <u>First press = HDMI output selection</u>: Press to select output/s from 1 to 8. Output(s) selected will be displayed on the Matrix display. Multiple outputs can be chosen. Then wait 3 seconds or press 'SELECT' to move to input selection. <u>Second Press = HDMI input selection</u>: Press to select the input from 1 - 8. The Input selected will be displayed on the Matrix display. Press the 'SELECT' button to confirm switch. Secondary Use: When the 'MENU' button has been pressed, buttons 3, 4, 7 & 8 are used as cursors to navigate the MENU system. 3 = Left, 4 = Up, 7 = Right, 8 = Down
4. 'MENU' Button	Press to access Matrix menu. Refer to full manual for details.
5. 'POWER' Button	Press to power on / off the Matrix
6. 'SELECT' Button	Press to confirm changes within the Matrix menu.
7. ESC	<u>Press to exit MENU mode Factory Reset</u> : The matrix can be factory reset by pressing and holding 'X' for 10 seconds, press the 'SELECT' button to confirm, or any other button to cancel.

Rear Panel Description



1. IR Control Input	3.5mm stereo connector to connect to Nice IR receiver for IR control of the Matrix
2. TCP/IP	RJ45 connector for TCP/IP and Web GUI control of the Matrix
3. RS-232	Phoenix connector for RS-232 control of the Matrix
4. EDID DIP switch	Used for global EDID settings
5. IR Outputs	3.5mm mono connector to connect to Nice IR emitter. Used for local source control
6. Optical and Analog Audio Outputs	Independent audio Matrix with 8 x dual outputs (Toslink + 3.5mm L/R line level stereo jack). For connection to 3rd party audio devices. NOTE: Analog audio outputs support 2ch PCM only
7. IR Inputs	3.5mm stereo connector to connect to Nice IR receiver or Control Processor. Used to extend IR from Matrix to HDBaseT Outputs 1-8.
8. Bi-directional RS-232 ports	Connect to third party control device to extend RS-232 commands to HDBaseT receivers RS-232 port
9. Optical (Toslink) Audio Input	Connect to source device for audio distribution within the EL-4KHDBT-RX-70-ARC-18G independent audio Matrix
10. Analog Audio Input	3.5mm L/R stereo jack. Connect to source device for audio distribution within the EL-4KHDBT-RX-70-ARC-18G independent audio Matrix
11. HDMI Inputs	Connect to source devices
12. HDMI Output	Connect to display device
13. HDBaseT Outputs	RJ45 HDBaseT port to connect to the HDBaseT input port of the compatible Nice HDBaseT receiver
14. IEC Power Socket	Use supplied IEC power cable
15. Power Switch	

Web GUI Control

The EL-4PKM-V88-A268-18G features an in-built Web GUI which can be used for control and configuration of the Matrix. By default the matrix is set to DHCP, however if a DHCP server (eg: network router) is not installed the matrix IP address will revert to below details:

Default Username: ELAN

Default Password: 3526

Video Port Default IP Address: 169.254.2.225

For further information please see the EL-4KPM-V88-A268-18G User Manual - available to download from the Nice website.

RS-232 Configuration

The RS-232 port is used for configuration and control of the product, as well as pass-through of RS-232 commands to a compatible Nice HDBaseT receiver / matrix device.

The default RS-232 communication settings are:

Baud Rate: 57600

Stop Bit: 1

Data Bit: 8

Parity Bit: none

For a complete RS-232 command list please see the EL-4KPM-V88-A268-18G User Manual - available to download from the Nice website.

EDID Control

EDID (Extended Display Identification Data) is a data structure that is used between a display and a source. This data is used by the source to find out what audio and video resolutions are supported by the display then from this information the source will discover what the best audio and video resolutions that need to be output.

While the objective of EDID is to make connecting a digital display to a source a simple plug and play procedure, issues do arise when multiple displays or video Matrix switching is introduced because of the increased number of variables.

By pre-determining the video resolution and audio format of the source and display device you can reduce the time need for EDID hand shaking thus making switching quicker, and more reliable.

Configuration of Matrix EDID settings can be achieved in one of three ways:

- 1. Using Matrix web browser interface
- 2. Using Matrix Front Panel (see below)
- 3. Using Matrix EDID dipswitches (see product User Manual)

To change the input signal type using the Matrix front panel press the following:

Using Matrix Front Panel Buttons

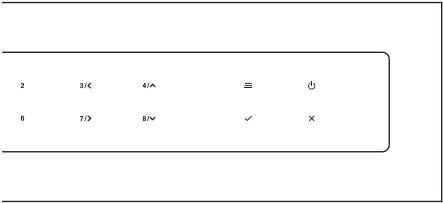
- a. Press the MENU button
- b. Scroll down to 'EDID Management'. Press the SELECT button to enter into EDID configuration mode
- c. Scroll through EDID management options and select 'Built-In EDID' or 'Copy Output xx'

Built-IN EDID Selection

- d. Select the input you wish to fix the EDID on (1-8) or select 'All'. Use UP/DOWN buttons to toggle the selection and press the SELECT button to move to EDID selection
- e. Select video resolution + audio format required (4K, 1080p, 3D, 2ch, 5.1 etc). Use UP/DOWN buttons to toggle the selection and press the SELECT button to apply. The SELECT button LED will flash orange when complete

Copy Output EDID

- f. Select the HDBaseT output you wish to copy the EDID from. Use UP/DOWN buttons to toggle the selection and press the SELECT button to move to the input selection that you wish to copy the EDID to
- g. Select the HDMI input (1-8) or select 'All' to copy to all HDMI inputs. Press the SELECT button to apply



Specifications

Video Input Connectors	8 x HDMI Type A, 19-pin, female
Video Output Connectors	1 x HDMI Type A, 19-pin, female, 8 x HDBaseT RJ45 connectors
Audio Input Connectors	1 x Analog audio L/R (3.5mm stereo Jack), 1 x Optical (S/PDIF)
Audio Output Connectors	8 x Analog audio L/R (3.5mm stereo Jack), 8 x Optical (S/PDIF)
RS-232 Serial Port	9 x 3-pin phoenix connector
IR Input Ports	9 x 3.5mm stereo jack
IR Output Ports	9 x 3.5mm mono jack
Rack Mountable	2U rack height, rack ears included
Casing Dimensions (W x H x D)	436mm x 88mm x 400mm (without feet)
Shipping Weight	10.5kg
Operating Temperature	23°F to 122°F (-5°C to +50°C)
Storage Temperature	-13°F to 158°F (-25°C to +70°C)
Power Supply	Internal 100-240V AC
HDBT Power Output	24V DC 0.75A Per Port

Package Contents

- 1 x EL-4KPM-V88-A268-18G
- 1 x Rack Mounting Kit
- 4 x Mounting feet and screws
- 1 x Remote Control
- 8 x IR Emitters
- 9 x IR Receivers
- 1 x Serial Cable - DB9 to 3-pin phoenix Connector
- 1 x IR Control Cable - 3.5mm to 3.5mm
- 1 x Quick Reference Guide
- IEC Power Cable(s) (US, UK, EU & AU)

NOTE: Battery not included. CR2025 battery required for remote control.

Information

- A.** Lisez ces instructions - Toutes les instructions de sécurité et d'utilisation doivent être lues avant d'utiliser ce produit.
- B.** L'appareil doit être raccordé à une prise de courant avec une mise à la terre de protection.
- C.** La prise de courant doit être installée à proximité de l'appareil et doit être facilement accessible.
- D.** Ne pas ingérer la batterie, risque de brûlure chimique
- E.** Ce produit contient une pile bouton. Si la pile bouton est avalée, elle peut provoquer de graves brûlures internes en seulement 2 heures et peut entraîner la mort.
- F.** Conservez les piles neuves et usagées hors de portée des enfants.
- G.** Si le compartiment à piles ne ferme pas correctement, cessez d'utiliser le produit et tenez-le hors de portée des enfants.
- H.** Si vous pensez que des piles ont pu être avalées ou placées à l'intérieur d'une partie du corps, consultez immédiatement un médecin.
- I.** **AVERTISSEMENT :** Avant d'installer ou d'utiliser l'appareil, veuillez vous référer aux informations figurant sur le boîtier inférieur extérieur pour les informations relatives à l'électricité et à la sécurité.

Example Schematic EL-4KPM-V88-A268-18G



Certifications

FCC Notice

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

CAUTION - changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

CANADA, INDUSTRY CANADA (IC) NOTICES

This Class B digital apparatus complies with Canadian ICES-003.

Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

CORRECT DISPOSAL OF THIS PRODUCT

This marking indicates that this product should not be disposed with other household wastes. To prevent possible harm to the environment or human health from uncontrolled waste disposal, recycle it responsibly to promote the sustainable reuse of material resources. To return your used device, please use the return and collection systems or contact the retailer where the product was purchased. They can take this product for environmentally safe recycling.

Technical Support:

800-421-1587

Technical Support Hours:

M – F, 6:00 a.m. – 4:00 p.m. PST

Nice North America

c/o Customer Service
5919 Sea Otter Place, Suite 100
Carlsbad, CA 92010

The logo for the company "Nice", consisting of the word "Nice" in a white, bold, sans-serif font, centered within a solid black square.

Niceforyou.com

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